CS24303 — THEORY (3.0 CR) MODULE V NOTES (10 TOPICS) BIT MESRA | NEP SCHEME

Module 5: Advanced Pattern Mining & Applications

Multilevel Rules, Antimonotonic Constraints, Colossal & Closed Patterns

Module V: Advanced Pattern Mining & Applications — 10 Topics Syllabus

1. Pattern Mining Road Map & Frontier Challenges
2. Multilevel Pattern Mining (Uniform vs Reduced Support, Redundancy)
3. Multidimensional Pattern Mining (Interdimension & Hybrid Rules)
4. Constraint-Based Frequent Pattern Mining (5 Classes & Mathematical Proofs)
5. Mining High-Dimensional Data & Top-k Patterns
6. Mining Colossal Patterns (Pattern-Fusion Algorithm & Core Patterns)
7. Mining Compressed Patterns (Closed vs Maximal Frequent Itemsets)
8. Mining Approximate & Noisy Patterns
9. Sequential & Graph Pattern Mining (GSP, PrefixSpan, gSpan)
10. Real-World Pattern Applications across 5 Key Industries

Topic 1: Advanced Pattern Mining Road Map

Advanced pattern mining extends standard single-dimensional market basket analysis into complex, multidimensional, constrained, and high-dimensional spaces:

1. **Complex Data Structures:** Moving from sets of items to sequential patterns (time-ordered sequences), structured subtrees, spatial topologies, and graph substructures.
2. **Multiple Abstraction Levels:** Mining across hierarchical taxonomy levels without generating millions of redundant rules.
3. **Multidimensional & Quantitative Predicates:** Handling relational tables with categorical and continuous numeric attributes simultaneously.
4. **Constraint-Based Pushing:** Pushing application-specific constraints into candidate generation and prefix tree construction to avoid combinatorial explosion.
5. **High-Dimensionality & Colossal Patterns:** Overcoming the 2^{100} search space barrier when patterns contain tens or hundreds of items.

Topic 2: Multilevel Pattern Mining

Items in real-world retail catalogs naturally form concept hierarchies (e.g., $\text{Computer} \rightarrow \text{Laptop} \rightarrow \text{Dell XPS 15}$).

Support Strategy	Mechanism & Threshold Settings	Key Tradeoffs
1. Uniform Support	The same minimum support threshold (e.g., $\text{min\_sup} = 5\%$) is applied at all abstraction levels.	Simple to implement; misses rare granular items at lower levels while producing trivial rules at top levels.
2. Reduced (Progressive) Support	Higher support thresholds at top levels ($\text{min\_sup} = 5\%$ for \`Computer\`), progressively lower thresholds at granular levels ($\text{min\_sup} = 0.5\%$ for \`Dell XPS 15\`).	Discovers rare low-level patterns; avoids flooding top levels with uninteresting rules.
3. Group-Based Support	Custom support thresholds set per product category (e.g., lower threshold for diamond rings, higher for milk).	Reflects business domain reality; requires domain expert configuration.

Redundant Rule Filtering:

A rule r_1 is redundant if it is an ancestor of rule r_2 across the hierarchy and its confidence is close to the expected value derived from r_2:

$$\text{Expected Confidence}(r_1) \approx \text{Confidence}(r_2)$$

Such ancestor rules convey no additional novel knowledge and are automatically filtered out.

Topic 3: Multidimensional & Quantitative Association Rules

Rule Paradigm	Definition & Predicate Structure	Example Rule
Single-Dimensional (Boolean)	Distinct predicates occur multiple times with different values (only one dimension `buys`).	$\text{buys}(X, \text{"Milk"}) \implies \text{buys}(X, \text{"Bread"})$
Interdimension Association Rule	Each predicate appears at most once in the rule across multiple distinct dimensions.	$\text{Age}(X, \text{"20..29"}) \wedge \text{Occupation}(X, \text{"Student"}) \implies \text{Buys}(X, \text{"Laptop"})$
Hybrid Association Rule	Combines multiple predicates where one or more predicates may appear repeatedly.	$\text{Age}(X, \text{"20..29"}) \wedge \text{Buys}(X, \text{"Laptop"}) \implies \text{Buys}(X, \text{"Backpack"})$

Quantitative Association Rules Mining Approaches:

1. **Static Discretization:** Partition continuous numeric attributes into predefined intervals (e.g., using equidepth binning) prior to mining.
2. **Dynamic Discretization (Grid-Based Clustering):** Discretizes quantitative dimensions dynamically during mining to maximize rule support and confidence (e.g., ARCS algorithm).

Topic 4: Constraint-Based Frequent Pattern Mining

Rather than mining unconstrained millions of patterns and filtering afterward, **Constraint-Based Mining** pushes user-specified constraints deeply into the mining algorithm (Apriori candidate generation and FP-Growth tree construction) to prune search branches early.

Constraint Class	Mathematical Property & Behavior	Concrete Example & Pushing Strategy
1. Antimonotonic	If an itemset S violates constraint C, all supersets $S' \supset S$ also violate C.	$\text{sum}(S.\text{price}) \leq 500, \min(S.\text{price}) \geq 20$. **Push:** Prune candidate branches immediately.
2. Monotonic	If an itemset S satisfies constraint C, all supersets $S' \supset S$ also satisfy C.	$\text{sum}(S.\text{price}) \geq 1000, \max(S.\text{price}) \geq 50$. **Push:** Once S satisfies C, skip testing all its supersets.
3. Succinct	All itemsets satisfying C can be explicitly enumerated using an analytical formula without candidate generation.	$\min(S.\text{price}) \leq 10, \text{Item.type} = \text{"Electronics"}$. **Push:** Filter items upfront before mining.
4. Convertible	A constraint that is neither monotonic nor antimonotonic, but converts into one when items are arranged in a specific sorting order.	$\text{avg}(S.\text{price}) \leq 50$ (Convertible antimonotonic if items are sorted by descending price).

🏛 Step-by-Step Formal Mathematical Proofs of Constraint Properties

Proof 1: Prove that $\text{sum}(S.\text{price}) \leq 500$ is Antimonotonic (assuming non-negative item prices $p_i \geq 0$).

Let S be an itemset violating the constraint $\implies \text{sum}(S.\text{price}) > 500$.
For any proper superset $S' \supset S, \text{sum}(S'.\text{price}) = \text{sum}(S.\text{price}) + \sum_{i \in S' \setminus S} p_i$.
Since $p_i \geq 0, \sum_{i \in S' \setminus S} p_i \geq 0 \implies \text{sum}(S'.\text{price}) \geq \text{sum}(S.\text{price}) > 500$.
Thus, every superset S' violates the constraint $\implies$ $\textbf{Antimonotonic Property Proven!}$

Proof 2: Prove that $\max(S.\text{price}) \geq 50$ is Monotonic.

Let S be an itemset satisfying the constraint $\implies \max(S.\text{price}) \geq 50$.
For any proper superset $S' \supset S, \max(S'.\text{price}) = \max(\max(S.\text{price}), \max((S' \setminus S).\text{price})) \geq \max(S.\text{price}) \geq 50$.
Thus, every superset S' automatically satisfies the constraint $\implies$ $\textbf{Monotonic Property Proven!}$

Topics 5 – 7: Mining High-Dimensional, Colossal & Compressed Patterns

1. Closed vs. Maximal Frequent Itemsets:

- **Closed Frequent Itemset:** An itemset X is closed if there exists no proper superset $Y \supset X$ such that $\text{Support}(Y) = \text{Support}(X)$. It provides a **lossless representation** of all frequent itemsets and their exact support counts.
- **Maximal Frequent Itemset (Max-Itemset):** An itemset X is maximal frequent if X is frequent and no proper superset $Y \supset X$ is frequent. It provides a compact **lossy representation** (the exact support of proper subsets is lost, only known to be $\geq \text{min\_sup}$).

🏛 Step-by-Step Solved Problem: Closed vs. Maximal Frequent Itemsets

Given Frequent Itemsets with Support Counts:

- $L_1 : \{A\} : 4,\ \{B\} : 4,\ \{C\} : 3,\ \{D\} : 2$
- $L_2 : \{A, B\} : 4,\ \{A, C\} : 2,\ \{B, C\} : 2$
- $L_3 : \{A, B, C\} : 2$

Analysis:

1. Itemset $\{A\}$ has support 4. Superset $\{A, B\}$ has support 4 $\implies$ $\{A\}$ is **NOT closed**.
2. Itemset $\{B\}$ has support 4. Superset $\{A, B\}$ has support 4 $\implies$ $\{B\}$ is **NOT closed**.
3. Itemset $\{A, B\}$ has support 4. Proper superset $\{A, B, C\}$ has support 2 (< 4) $\implies$ $\{\mathbf{A}, \mathbf{B}\}$ is **CLOSED**.
4. Itemset $\{A, B, C\}$ has support 2. No frequent superset exists $\implies$ $\{\mathbf{A}, \mathbf{B}, \mathbf{C}\}$ is **CLOSED and MAXIMAL**.
5. Itemset $\{D\}$ has support 2. No frequent superset exists $\implies$ $\{\mathbf{D}\}$ is **CLOSED and MAXIMAL**.

2. Mining Colossal Patterns via Pattern-Fusion:

When datasets contain colossal patterns (e.g., frequent itemsets of length 50 to 100 items), traditional level-wise and pattern-growth methods collapse because exploring the 2^{100} search space is computationally intractable. **Pattern-Fusion** jumps across the search space by fusing smaller core frequent patterns into large colossal patterns in single leaps rather than incremental 1-item extensions.

Topic 9: Sequential & Graph Pattern Mining (PrefixSpan & gSpan)

1. Sequential Pattern Mining (PrefixSpan):

A **Sequence Database** stores ordered lists of itemsets associated with timestamps (e.g., customer purchase sequences: $\langle (a)(abc)(df)(d) \rangle$).

- **PrefixSpan (Prefix-projected Sequential Pattern Mining):** Mines sequences without candidate generation by recursively projecting sequence databases into smaller projected databases based on frequent prefixes.
- Avoids candidate generation and exhibits linear scalability with respect to database size.

2. Graph Pattern Mining (gSpan):

Discovers frequent subgraphs in chemical compounds, biological networks, and social graphs. **gSpan** uses a canonical **DFS Code Dictionary** to uniquely identify and order graphs, pruning isomorphic duplicate candidates without expensive subgraph isomorphism testing.

Topic 10: Real-World Industrial Applications

1. **Retail & E-Commerce:** Market basket cross-selling, automated recommendation systems ("Customers who bought this also bought..."), promotional bundle optimization.
2. **Web Analytics & Clickstream Mining:** Discovering common web navigation paths to optimize website UI layouts and personalize dynamic landing pages.
3. **Healthcare & Bioinformatics:** Mining co-occurring genetic mutations, drug-drug interaction contraindications, diagnostic clinical pathways.
4. **Software Engineering & Bug Localization:** Mining execution traces of passing vs. failing test cases to identify fault-inducing code dependencies.
5. **Financial Fraud Detection:** Discovering synchronized multi-account transaction patterns characteristic of money laundering.

🏛️ Top BIT Mesra Exam Questions & Answers (Module V)

Q1. Prove why the constraint $\text{sum}(S.\text{price}) \leq 500$ is Antimonotonic assuming non-negative prices. (6 Marks)

Proof: Let S be an itemset such that $\text{sum}(S.\text{price}) > 500$ (violating the constraint). For any superset $S' \supset S$, since item prices are non-negative ($p_i \geq 0$), $\text{sum}(S'.\text{price}) = \text{sum}(S.\text{price}) + \sum_{i \in S' \setminus S} p_i \geq \text{sum}(S.\text{price}) > 500$. Therefore, every superset S' is guaranteed to violate the constraint. By definition, $\text{sum}(S.\text{price}) \leq 500$ is **Antimonotonic**.

Q2. Explain the core mechanism of Pattern-Fusion for mining colossal patterns. (8 Marks)

Answer: Colossal patterns are massive patterns (length ≥ 50) that cannot be mined by traditional bottom-up $k \to k+1$ extensions due to the combinatorial explosion of 2^{50} sub-patterns. Pattern-Fusion samples small, bounded *core patterns*, computes their equivalence classes, and leaps directly across the lattice by fusing multiple disjoint core patterns into colossal patterns in single steps, bypassing exponential intermediate levels.

Q3. Why is mining Closed Frequent Itemsets preferred over mining all frequent itemsets? (6 Marks)

Answer: If an itemset has 100 items, it contains $2^{100} - 1 \approx 10^{30}$ frequent sub-itemsets. Storing and mining all of them causes combinatorial explosion. The set of **Closed Frequent Itemsets** contains only those itemsets with no superset having the same support. It provides a 100% **lossless compression** of the frequent pattern space: the exact support of every single frequent itemset can be derived directly from the closed itemsets with zero information loss, while dramatically reducing memory consumption and mining runtime.

Q4. Differentiate between PrefixSpan and GSP algorithms for sequential pattern mining. (6 Marks)

GSP (Generalized Sequential Patterns): An Apriori-like level-wise algorithm that generates candidate sequences C_k from L_{k-1} and makes repeated full scans of the sequence database.
PrefixSpan (Prefix-projected Pattern Growth): A pattern-growth method that projects the sequence database into smaller suffix databases based on frequent prefixes, mining sequential patterns recursively without candidate generation.

Q5. Explain how quantitative association rules are mined using grid-based clustering. (6 Marks)

Answer: In grid-based clustering (e.g., ARCS system), continuous numeric dimensions (e.g., Age and Salary) are partitioned into 2D grid cells. Neighboring grid cells with high data density that satisfy minimum support are dynamically clustered into convex bounding polygons. Association rules are then extracted from these clustered regions, avoiding artificial rigid boundary partitions produced by standard static binning.

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